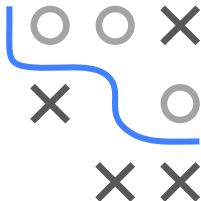


ONLINE CONVEX OPTIMIZATION

- One of the most relevant instantiations of the online learning problem is the problem of *online convex optimization* (OCO), which is characterized by a loss function

$$\ell(a, z) : \mathcal{A} \times \mathcal{Z} \rightarrow \mathbb{R},$$

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- Note that both OLO and OQO belong to the class of online convex optimization problems:

- Online linear optimization (OLO) with convex action spaces:*

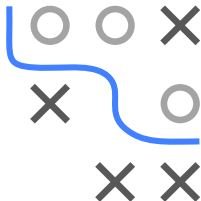
$$\ell(a, z) = a^\top z$$

is a convex function in $a \in \mathcal{A}$, provided $\mathcal{A}$ is convex.

- Online quadratic optimization (OQO) with convex action spaces:*

$$\ell(a, z) = \frac{1}{2} \|a - z\|_2^2$$

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ONLINE GRADIENT DESCENT: MOTIVATION

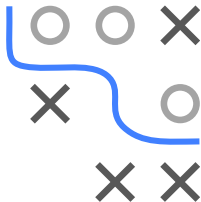
- We have seen that the FTRL algorithm with the ℓ_2 norm regularization $\psi(a) = \frac{1}{2\eta} \|a\|_2^2$ achieves satisfactory results for online linear optimization (OLO) problems, that is, if $(a, z) = L^{\text{lin}}(a, z) := a^\top z$, then we have

- *Fast updates* — If $\mathcal{A} = \mathbb{R}^d$, then

$$a_{t+1}^{\text{FTRL}} = a_t^{\text{FTRL}} - \eta z_t, \quad t = 1, \dots, T;$$

- *Regret bounds* — By an appropriate choice of η and some (mild) assumptions on $\mathcal{A}$ and $\mathcal{Z}$, we have

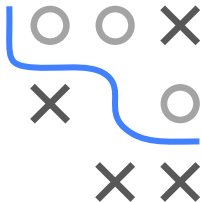
$$R_T^{\text{FTRL}} = o(T).$$



ONLINE GRADIENT DESCENT: MOTIVATION

Apparently, the nice form of the loss function L^{lin} is responsible for the appealing properties of FTRL in this case. Indeed, since $\nabla_a L^{\text{lin}}(a, z) = z$ note that the update rule can be written as

$$\mathbf{a}_{t+1}^{\text{FTRL}} = \mathbf{a}_t^{\text{FTRL}} - \eta \mathbf{z}_t = \mathbf{a}_t^{\text{FTRL}} - \eta \nabla_a L^{\text{lin}}(\mathbf{a}_t^{\text{FTRL}}, \mathbf{z}_t).$$

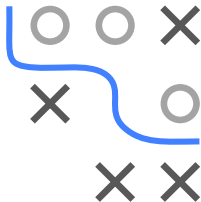


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Interpretation: In each time step $t + 1$, we are following the direction with the steepest decrease of the most recent loss (represented by $-\nabla L^{\text{lin}}(\bar{a}_t^{\text{FTRL}}, z_t)$) from the current "position" $\bar{a}_t^{\text{FTRL}}$ with the step size η

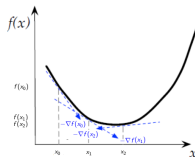


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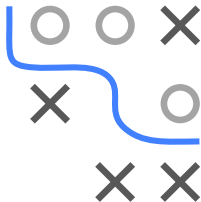
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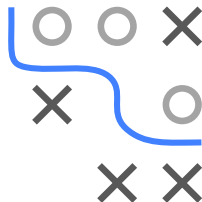


⇒ Gradient Descent.



ONLINE GRADIENT DESCENT: MOTIVATION

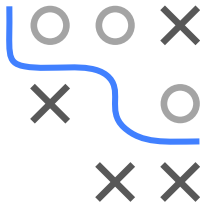
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$$\begin{aligned} f : S \rightarrow \mathbb{R} \text{ is convex} &\Leftrightarrow f(y) \geq f(x) + (y - x)^\top \nabla f(x) \text{ for any } x, y \in S \\ &\Leftrightarrow f(x) - f(y) \leq (x - y)^\top \nabla f(x) \text{ for any } x, y \in S. \end{aligned}$$



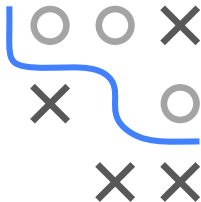
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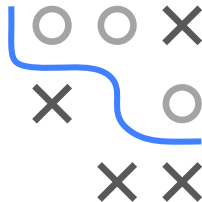
- This means if we are dealing with a loss function $\ell : \mathcal{A} \times \mathcal{Z} \rightarrow \mathbb{R}$, which is convex and differentiable in its first argument ($\mathcal{A}$ has also to be convex), then

$$(a, z) - (\tilde{a}, z) \leq (a - \tilde{a})^\top \nabla_a \ell(a, z), \quad \forall a, \tilde{a} \in \mathcal{A}, z \in \mathcal{Z}.$$



ONLINE GRADIENT DESCENT: MOTIVATION

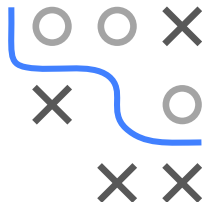
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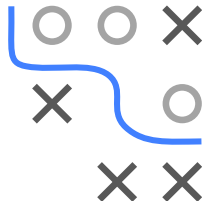


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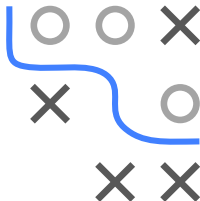
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Conclusion: The regret of a learner with respect to a differentiable and convex loss function is bounded by the regret corresponding to an online linear optimization problem with environmental data $\tilde{z}_t = \nabla_a(a_t, z_t)$.



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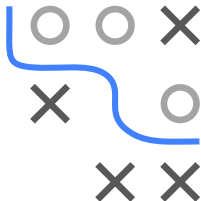
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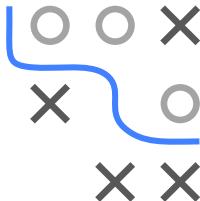
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- $\leadsto$ Incorporate the substitution $\tilde{z}_t = \nabla_a(a_t, z_t)$ into the update formula of FTRL with squared L2-norm regularization.



ONLINE GRADIENT DESCENT: DEFINITION

- The corresponding algorithm which chooses its action according to these considerations is called the *Online Gradient Descent* (OGD) algorithm with step size $\eta > 0$. It holds in particular,

$$a_{t+1}^{\text{OGD}} = a_t^{\text{OGD}} - \eta \nabla_a(a_t^{\text{OGD}}, z_t), \quad t = 1, \dots, T. \quad (1)$$

(Technical side note: For this update formula we assume that $\mathcal{A} = \mathbb{R}^d$. Moreover, the first action a_1^{OGD} is arbitrary.)

